

Chance Krenzer

Los Angeles, California | 818-268-7308 | US Citizen (ITAR Eligible)
chance.krenzer@gmail.com | [linkedin.com/in/chance-krenzer](https://www.linkedin.com/in/chance-krenzer) | chancekrenzer.com

EDUCATION

Case Western Reserve University, Class of 2029

BS, Mechanical Engineering | BS, Aerospace Engineering

Cleveland, Ohio

Dean's High Honors List, Cumulative GPA: 3.81/4.00

Harvard-Westlake School, Class of 2025

National Merit Scholar, Dean's List, Captain – Varsity Track Team, Presidential Gold Volunteer Service Award (2023)

Los Angeles, California

TECHNICAL SKILLS

CAD & Simulation – SolidWorks (Parametric Design, PDM, Finite Element Analysis (FEA), Flow Simulation/CFD), 3D Printing
Ansys Fluent, Mastercam 2026

Aerospace – UAV/VTOL Airframe Design, Aerodynamics, Propulsion Selection, Flight Controls (PID), Avionics

Programming – Python, C, Java, MATLAB, Arduino, Embedded Systems, PID Control, PCB Design (learning)

Grid & Communication Protocols – Modbus, IEEE 2030.5, Distributed Energy Resource Management Systems (DERMS)

Manufacturing – CNC Waterjet, Mill, Lathe, MIG & TIG Welding, Laser Cutter, Soldering

Software – Microsoft Excel, Word, PowerPoint, Outlook

Relevant Coursework - Mechanical Manufacturing, Thermodynamics, Dynamics, Computers, Strength of Materials

PROFESSIONAL EXPERIENCE

Los Angeles Department of Water and Power (LADWP)

Los Angeles, California

Distributed Energy Resource Management Systems (DERMS) Fellowship

May 2026 – July 2026

- Analyzed operational gaps created by the growing influx of distributed energy resources (DERs) and designed DERMS solutions to help grid operators monitor and manage DER integration.
- Specialized in DERMS software platforms and utility communication protocols, working with Modbus and IEEE 2030.5 to evaluate telemetry, monitoring, and control pathways between the utility and DER endpoints.
- Collaborated with a cross-functional team to synthesize technical findings into a summary presentation for stakeholders.

Dronefly

Los Angeles, California

Drone Repair & Operations Intern

May 2026 – August 2026

- Diagnosed and serviced high-end commercial drones returned through warranty claims, maximizing successful repairs to preserve profit margin on each returned unit.
- Built a new, more comprehensive file management system for warranty and repair records, increasing business-operations efficiency and traceability.

EXTRACURRICULAR INVOLVEMENT

CWRU Vertical Take-Off and Landing (VTOL) Team (1st Place, 2025 Championship)

Cleveland, Ohio

Mechanical Team Specialist

August 2025 – Present

- Designed and fabricated a payload pickup-and-release mechanism tolerant of positional and alignment error, enabling reliable autonomous operation in competition.
- Collaborated with the team to CAD and fabricate a jig to precisely epoxy an aluminum motor mount to a carbon fiber arm for an X8 configuration of the competition drone.
- Manufactured aluminum motor mount components using the CNC waterjet, belt sander, and mill.

Mechanical Design Lead – High-Speed Quadcopter

- Led a team of engineers developing a high-speed quadcopter targeting 500 kph; designed the frame parametrically and ran flow simulations (CFD) to minimize drag while retaining high-speed structural strength.
- Specified the electronics and mounting architecture and submitted a bill of materials (BoM) to secure team funding.

CWRU Varsity Cross Country / Track and Field

August 2025 – Present

PROJECTS

ESP32 Drone – Personal Project

December 2025 – February 2026

- Researched and sourced motors, sensors, and components; designed wiring diagrams for a custom motor driver and inertial measurement unit (IMU), and am learning PCB design to consolidate components onto a single circuit board.
- Designed and 3D-printed a PETG frame in CAD, validated with SolidWorks FEA to withstand motor-induced loads while maintaining a 2:1 thrust-to-weight ratio.
- Wrote and tuned custom flight-control software in Arduino using a PID controller.

CWRU Hackathon – 4th Place

November 2025

- Built a robotic hand controlled wirelessly by electromyography (EMG) and wrote the code for both the transmitter and receiver.
- Modeled the fingers and palm in SolidWorks, using mechanical mates to simulate articulation before 3D printing.